

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations
 May 2025

MA2005-1 – LINEAR ALGEBRA AND ITS APPLICATIONS

(For CS, IS, AM, CC,CB,RI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. System of equations $AX=B$ has infinite number of solutions if
 - A) $\text{Rank}[A] = \text{Rank}[A|B] = \text{number of unknowns}$
 - B) $\text{Rank}[A] = \text{Rank}[A|B] < \text{number of unknowns}$
 - C) $\text{Rank}[A] \neq \text{Rank}[A|B]$
 - D) $\text{Rank}[A] = \text{Rank}[A|B] > \text{number of unknowns}$
2. If $A = \begin{pmatrix} -1 & 1 \\ 2 & 4 \end{pmatrix}$ then $\text{tr}(AA^T)$ is
 - A) 21
 - B) 22
 - C) 3
 - D) 26
3. The eigen values of $A = \begin{bmatrix} 3 & 2 \\ 3 & 8 \end{bmatrix}$ are
 - A) 7,4
 - B) 9,2
 - C) 8,3
 - D) 6,5
4. If A is 3×4 matrix the maximum possible value of the rank A is
 - A) 4
 - B) 3
 - C) 2
 - D) 0
5. The characteristic equation of the matrix $A = \begin{bmatrix} 2 & 0 \\ 4 & 7 \end{bmatrix}$ is:
 - A) $\lambda^2 - 9\lambda + 14 = 0$
 - B) $\lambda^2 - 14\lambda + 9 = 0$
 - C) $\lambda^2 - 9\lambda - 14 = 0$
 - D) $\lambda^2 - 14\lambda - 9 = 0$
6. Number of pivot columns in the matrix $\begin{bmatrix} 0 & 0 & 1 & 0 \\ 2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$ is:
 - A) 1
 - B) 2
 - C) 3
 - D) 4
7. Which of the following method is not used to find the solution of the system of equations:
 - A) Gauss elimination method
 - B) Gauss Seidel method
 - C) Rayleigh's power method
 - D) LU decomposition method
8. If the LU factorization of a matrix A is $A=LU$ where $A = \begin{bmatrix} 2 & 5 \\ -3 & -4 \end{bmatrix}$ then U is
 - A) $\begin{bmatrix} 2 & 5 \\ 0 & 7 \end{bmatrix}$
 - B) $\begin{bmatrix} 1 & 4 \\ -5 & 1 \end{bmatrix}$
 - C) $\begin{bmatrix} 2 & 5 \\ 1 & 7 \end{bmatrix}$
 - D) $\begin{bmatrix} 1 & 4 \\ -7/2 & 1 \end{bmatrix}$
9. If $\{u_1, u_2, \dots, u_n\}$ is linearly independent set, then $c_1u_1 + c_2u_2 + \dots + c_nu_n = 0$ has
 - A) infinitely many solutions
 - B) unique nonzero solution
 - C) unique zero solution
 - D) both B and C are possible

10. The row space of an $m \times n$ matrix A is a subspace of
 A) R^n B) A
 C) R^m D) R
11. If $S = \{v_1, v_2, \dots, v_n\}$ is a set of vectors in a finite dimension vector space V , then S is called a basis for V if
 A) S spans V B) S is linearly independent
 C) either (A) or (B) D) both (A) and (B)
12. Let $T: R^2 \rightarrow R^2$ be the linear transformation defined by $T(x, y) = (0, y)$. Then matrix of linear transformation with respect to standard basis is
 A) $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ B) $\begin{bmatrix} 0 & -1 \\ 0 & -1 \end{bmatrix}$
 C) $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ D) $\begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix}$
13. The set of all solution to the homogeneous equation system $AX = 0$, where A be an $m \times n$ matrix is
 A) Null space of A B) Column space of A
 C) Row space of A D) Null space of A and column space of A
14. If a basis of a vector space has 3 elements, then dimension of the vector space is _____
 A) less than 4 B) greater than 3
 C) 3 D) 0
15. If $u = \begin{bmatrix} 2 \\ 3 \\ -5 \end{bmatrix}$, $v = \begin{bmatrix} 2 \\ 0 \\ -3 \end{bmatrix}$ then inner product of u and v is
 A) -5 B) -11
 C) 19 D) 11
16. Let $T: R^3 \rightarrow R^3$ be the linear transformation with nullity of T as 1 then rank of T is
 A) 1 B) 2
 C) 3 D) 4
17. In PCA, what is the relationship between the first principal component and the second principal component
 A) They are orthogonal to each other B) They are positively correlated
 C) They are negatively correlated D) There is no defined relationship between them
18. The matrix of the quadratic form $4x_1x_2 + 6x_1x_3 - 8x_2x_3$
 A) $\begin{bmatrix} 0 & 2 & 2 \\ 2 & 0 & -4 \\ 2 & -4 & 0 \end{bmatrix}$ B) $\begin{bmatrix} 0 & 1 & 3 \\ 2 & 0 & -4 \\ 3 & -4 & 0 \end{bmatrix}$
 C) $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & -4 \\ 3 & -4 & 1 \end{bmatrix}$ D) $\begin{bmatrix} 0 & 2 & 3 \\ 2 & 0 & -4 \\ 3 & -4 & 0 \end{bmatrix}$
19. Let B be the mean deviation form of N observation vector then covariance matrix is
 A) $S = \frac{1}{N-1} B^T$ B) $S = \frac{1}{N-1} B B^T$
 C) $S = \frac{1}{N+1} B B^T$ D) $S = \frac{1}{N} B B^T$
20. Singular values of $A = \begin{bmatrix} 1 & 0 \\ 0 & -3 \end{bmatrix}$ are
 A) 1, -3 B) 9, 1
 C) $\sqrt{3}$, 1 D) 3, 1

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit - I		Marks	BT*	CO*	PO*
1. a)	Diagonalize the matrix $A = \begin{bmatrix} -1 & 2 \\ 2 & -1 \end{bmatrix}$. Hence find A^6 .	6	L*3	2	2
b)	Use Gauss-Seidel iteration method to solve the following system of linear equations $20x + y - 2z = 17$, $3x + 20y - z = -18$, $2x - 3y + 20z = 25$, start with $x^{(0)} = y^{(0)} = z^{(0)} = 0$ and carry out three iterations.	6	L3	1	2
c)	Find the general solution of the system whose augmented matrix is $\begin{bmatrix} 1 & 3 & 4 & 7 \\ 3 & 9 & 7 & 6 \end{bmatrix}$	4	L3	1	2
2. a)	Find the value of a and b so that the equations $2x + 3y + 5z = 9$, $7x + 3y - 2z = 8$, $2x + 3y + az = b$ has i) no solution ii) a unique solution iii) more than one solution.	6	L3	1	2
b)	Solve the system of equations $x + y + z = 1$, $4x + 3y - z = 6$, $3x + 5y + 3z = 4$ by LU factorization method.	6	L3	1	2
c)	Using the power method, find the dominant eigen value and corresponding eigen vector of the matrix $A = \begin{bmatrix} 1 & -3 & 2 \\ 4 & 4 & -1 \\ 6 & 3 & 5 \end{bmatrix}$ starting with the initial value $[1 \ 0 \ 0]^T$. Carry out four iterations.	4	L3	2	2
3. a)	Find all the eigen values of $A = \begin{bmatrix} 13 & -4 & 2 \\ -4 & 13 & -2 \\ 2 & -2 & 10 \end{bmatrix}$. Find an eigen vector corresponding to the largest eigen value.	6	L3	1	2
b)	Reduce the given matrix A to echelon form and find its rank				
	$A = \begin{bmatrix} 0 & -3 & -6 & 4 & 9 \\ -1 & -2 & -1 & 3 & 1 \\ -2 & -3 & 0 & 3 & -1 \\ 1 & 4 & 5 & -9 & -7 \end{bmatrix}$	6	L3	1	2
c)	Define Trace of a matrix. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 5 & 2 \\ -1 & 3 \end{bmatrix}$ then verify that i) $tr(A + B) = tr(A) + tr(B)$ ii) $tr(AB) = tr(BA)$	4	L3	2	2
Unit - II					
4. a)	Let $T: R^3 \rightarrow R^2$ be a linear map defined by $T(x, y, z) = (y - x, y - z)$. Find the null space and range space of T. Also find dimension of null space and range space of T.	6	L3	4	2
b)	Find a basis for the row space of				
	$\begin{bmatrix} 1 & -3 & 4 & -1 & 9 \\ -2 & 6 & -6 & -1 & -10 \\ -3 & 9 & -6 & -6 & -3 \\ 3 & -9 & 4 & 9 & 0 \end{bmatrix}$	6	L3	3	2
c)	Check whether the set of vectors $\{(1, 2, 3), (4, 5, 6), (2, 1, 0)\}$ is linearly dependent or linearly independent.	4	L3	3	2

5. a) Prove that a nonempty subset W of a vector space V over the field F is a subspace if and only if for any $\alpha, \beta \in W$ and $c \in F$, $c\alpha + \beta \in W$.
- b) Let $W = \text{span}\{x_1, x_2, x_3\}$ where $x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, $x_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$, $x_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$.
Construct an orthogonal basis for W using Gram-Schmidt process.
- c) Find the co-ordinate vector of $(2, 7, -4)$ in R^3 with respect to the basis $\{(1, 2, 0), (1, 3, 2), (0, 1, 3)\}$.
6. a) Determine whether $(1, 1, 1, 1)$, $(1, 2, 3, 2)$, $(2, 5, 6, 4)$, $(2, 6, 8, 5)$ form a basis of R^4 . If not find the dimension of the subspace they span.
- b) Prove that a set of vector $\{v_1, v_2, \dots, v_n\}$ is linearly dependent if and only if at least one of them is linear combination of the others.
- c) Find the matrix of linear transformation $T: V_2(R) \rightarrow V_3(R)$ defined by $T(x, y) = (x + y, x, 3x - y)$ with respect to the bases $B_1 = \{(1, 1), (3, 1)\}$ and $B_2 = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}$.

6	L1	3	1
6	L3	4	2
4	L3	3	2
6	L3	3	2
6	L1	3	1
4	L3	4	2

Unit – III

7. a) Find QR factorization of matrix $A = \begin{bmatrix} 1 & 3 & 5 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 3 & 3 \end{bmatrix}$
- b) Find a least square solution of $AX=B$ for $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 \\ 3 \\ 8 \\ 2 \end{bmatrix}$
8. a) Find a singular value decomposition of matrix $A = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}$
- b) Convert the matrix $\begin{bmatrix} 1 & 5 & 2 & 6 & 7 & 3 \\ 3 & 11 & 6 & 8 & 15 & 11 \end{bmatrix}$ of observations to mean deviation form and construct the sample covariance matrix.

8	L3	5	2
8	L3	5	2
10	L3	5	2
6	L3	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
